

# Amazon Sales Analysis Project

## 1.Dataset Overview

- Dataset Name: **Amazon Sale Report.csv**
- Total Records (Rows): **128,975**
- Total Columns: **24**

## 2.Data Types Identified

The dataset contains the following data types:

- **Object:** 19 columns
- **Integer (int64):** 2 columns
- **Float (float64):** 2 columns
- **Boolean:** 1 column

The majority of the columns contain categorical or text information such as Order ID, Category, Status, SKU, City, and State.

## 3.Duplicates

- I have found **8597** duplicate Order ID .
- But this is possible because some one order multiple items like kurta ,jeans ,shirt etc. Amazon will store this in three different rows with same order ID.

## 4.Missing values

- **Investigation 1: Amount and Currency**

### **Business Insight:**

The missing values in the Amount and Currency columns are strongly associated with cancelled orders.

### **Hypothesis:**

Cancelled orders may not have completed the payment process or their transaction details were not recorded. Additional payment-related information would be needed to confirm this.

- **Investigation 2: Missing Amount by Order Status**

**Observation:**

A total of **7,795** records have missing Amount values.

After grouping by **Status**, it was found that:

**7,566** missing values belong to **Cancelled** orders.

Only **229** missing values belong to all other statuses combined.

**Business Insight:**

Missing Amount values are **highly concentrated** in cancelled orders and are **not randomly distributed** across the dataset.

- **Investigation 4:**

**Observation:**

The crosstab analysis shows that Amazon fulfilment has a higher number of cancelled and pending orders than Merchant fulfilment.

However, the shipping-related statuses differ between Amazon and Merchant. Amazon mainly uses a single "**Shipped**" status, whereas Merchant uses more detailed shipping statuses such as "**Delivered to Buyer**," "**Picked Up**," and "**Returned to Seller**."

**Business Insights:**

The fulfilment workflow differs between Amazon and Merchant, so direct comparison of individual status counts should be done carefully.

Cancelled and Pending orders appear to be the most significant statuses for further investigation.

**Note:** Instead of calculation numbers of orders cancelled and pending, we have to find cancellation rate% to make fair comparison.

## 5. Exploratory Data Analysis (EDA)

### ❖ Cancellation Analysis

- **Overall Cancellation Rate:**

**Business Question:** What is the overall rate of cancellation of orders?

**Observation:**

- Total Orders: 128,975
- Cancelled orders: 18332
- Cancellation Rate: 14.21%

**Business Insights:**

Approximately **14 out of every 100 orders** are cancelled before completion.

This indicates that a noticeable proportion of customer orders do not reach successful completion.

## 1. Cancellation Rate by Fulfilment

**Business Question:** Which fulfilment type (Amazon or Merchant) has a higher cancellation rate?

**Observation:**

- Amazon handled **89,698** orders, while Merchant handled **39,277** orders.
- Although Amazon had a higher number of cancelled orders (**11,471**), it also processed a much larger number of total orders.
- Merchant Fulfilment has the **higher cancellation rate (17.47%)** compared to Amazon Fulfilment (**12.79%**).

**Business Insight:**

- Comparing only the number of cancelled orders can be misleading because Amazon processes significantly more orders.
- Cancellation rate provides a more accurate measure of fulfilment performance.
- Merchant fulfilment experiences cancellations more frequently relative to the number of orders it processes.

## 2. State wise Cancellation Rate

**Business Question:** Which State have highest rate of cancellation?

**Observations:**

- Himachal Pradesh recorded the highest meaningful cancellation rate (18.53%).
- Mizoram and Kerala also showed relatively high cancellation rates.
- States such as Karnataka, Maharashtra, Delhi, and Tamil Nadu have much larger order volumes but comparatively lower cancellation rates.

- APO (100%) and Lakshadweep (25%) have very small sample sizes, making them unsuitable for business conclusions.

#### **Business Insights:**

Order cancellation behavior varies across geographical regions, indicating that regional operational factors may influence cancellation patterns.

### **3.Cancellation Data & Courier Status Investigation**

#### **(Root Cause Investigation)**

#### **Missing Amount & Currency**

- **Observation:** Column containing Amount and currency have exactly **7795** missing values  
Of the 7,795 records with missing Amount values, 7,566 were associated with Cancelled orders, indicating a strong association between missing transaction details and order cancellation.

#### **Courier Status**

**Business Question:** Does the Courier status have the any relations with order cancellations

#### **Observation**

- 5,935 orders have a courier status of Cancelled.
- Most records with **Courier Status = Cancelled** also have **Order Status = Cancelled (5,840 orders)**.
- A large number of cancelled orders (**5,631**) belong to the **Unshipped** courier status.
- A very small number of records (93) show **Courier Status = Cancelled** while **Order Status = Shipped**, which may indicate data inconsistency or timing differences in status updates.

#### **Business Insight:**

Most order cancellations occur **before the product enters the shipping process**. Based on the available data, it is not appropriate to conclude that courier services are the primary cause of cancellations. Further investigation into customer behavior, inventory availability, or order processing is required.

## **4.Product category Analysis**

**Business Question: Which product category generates the highest revenue?**

**Observation:**

- The Set category generates the highest revenue (₹39.20 million).
- Kurta is the second highest revenue-generating category.
- Western Dress ranks third.
- Categories such as Dupatta, Saree, and Bottom contribute comparatively lower revenue.

**Business Insights:**

- The Set category is the company's most valuable product category in terms of revenue.
- Further analysis of quantity sold and average selling price shows that the Set category's strong revenue performance is supported by both sales volume and its relatively high average selling price.

**Business Question: Which product categories sell the highest number of units?**

**Observation:**

- The Set category recorded the highest quantity sold (45,289 units).
- Kurta ranked second with 45,045 units, which is very close to the Set category.
- Western Dress and Top followed with considerably lower sales quantities.
- Other categories contributed relatively small sales volumes.

**Business Insights:**

- Set and Kurta have nearly identical sales volumes.
- However, the Set category generates significantly higher revenue than Kurta.

- This suggests that Set products are sold at a higher average selling price.

**Observation:**

- Set has an ASP of about ₹865.64, while Kurta has an ASP of about ₹472.85. Yet their quantities sold were almost identical:
- Set → **45,289** units sold
- Kurta → **45,045** units sold

**Strong Insight:**

**Set generates much higher revenue primarily because its average selling price is substantially higher than Kurta, while both categories have almost the same sales volume.**

## **5. Geographic Analysis**

### **Investigation: State wise revenue analysis**

**Observation:**

- Maharashtra generates the highest revenue of ₹13.34M.
- Karnataka ranks second with revenue of ₹10.48M.
- The top 5 states together generate approximately ₹44.6M revenue.
- Smaller states such as Lakshadweep, Ladakh, and Mizoram contribute comparatively low revenue.

**Insights:**

- Maharashtra and Karnataka are the strongest revenue markets.
- Revenue is significantly concentrated in a few major states.
- High revenue may be due to higher order volume or higher spending per order.

### **Investigation: City-wise Analysis**

**Observation:**

- Bengaluru generated the highest revenue of ₹80.83 lakh.
- Hyderabad ranked second with ₹56.00 lakh.
- Mumbai and Delhi generated ₹42.93 lakh and ₹41.70 lakh respectively.
- Major metropolitan cities dominate the top revenue-generating cities.

**Insights:**

- Bengaluru is the highest revenue-generating city in the dataset.
- Revenue is concentrated among a few major cities.

## 6.Monthly Trend Analysis First

**Investigation: Monthly Revenue Analysis**

**Observations:**

- March 2022 recorded revenue of ₹1,01,683.85, but this represents only one day of data.
- April 2022 generated the highest recorded revenue of ₹2,88,38,708.32.
- May 2022 generated ₹2,62,26,476.75.
- June 2022 generated ₹2,34,25,809.38, with data available only up to June 29.
- Revenue declined from April through June.

**Insights:**

April 2022 recorded the highest monthly revenue of approximately ₹2.88 crore. Recorded revenue declined in May and June; however, June represents a partial month and should therefore be interpreted cautiously. The complete-month comparison between April and May shows a decline in recorded revenue.

**Investigation: Monthly distribution of product quantities**

**Observation:**

- March 2022 recorded 156 units.
- April 2022 recorded the highest quantity with 44,206 units.
- May 2022 recorded 38,011 units.
- June 2022 recorded 34,276 units.
- Quantity declined from April through June.

**Insights:** April recorded the highest sales quantity, followed by a decline in May and June. However, March and June represent partial periods because the dataset covers only March 31 and up to June 29 respectively. Therefore, monthly quantity comparisons should consider the different lengths of the available data periods.

### **Investigation: Monthly ASP**

#### **Observations:**

- March ASP: ₹651.82
- April ASP: ₹652.37
- May ASP: ₹689.97
- June ASP: ₹683.45
- ASP increased significantly from April to May and remained relatively high in June.

**Insights:** The average selling price increased from ₹652.37 in April to ₹689.97 in May, despite a decline in both quantity and total revenue. This indicates that the decline in revenue was primarily associated with lower sales volume rather than a reduction in average selling price.

### **Investigation: Month\_over\_Month revenue Change**

#### **Observation:**

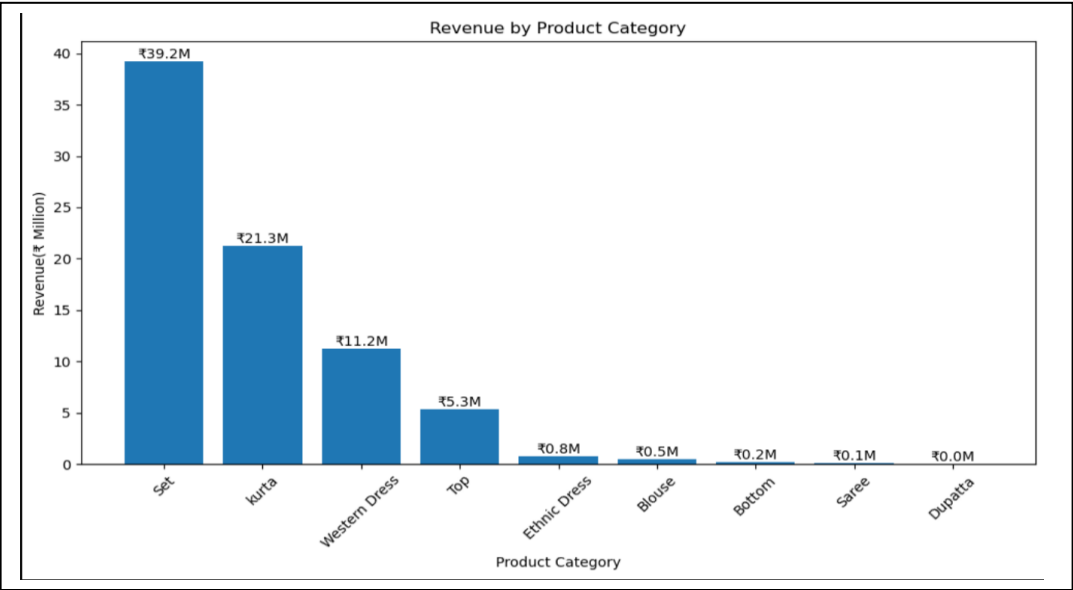
- April shows a very large increase of 28,261.15% compared with March.
- This extreme increase is primarily caused by March containing only one day of data.
- Revenue decreased by 9.06% from April to May.
- Revenue decreased by 10.68% from May to June.

#### **Insights:**

- **"Revenue declined by 9.06% from April to May. The apparent 28,261.15% increase from March to April was excluded from trend interpretation because March contains only one day of data."**



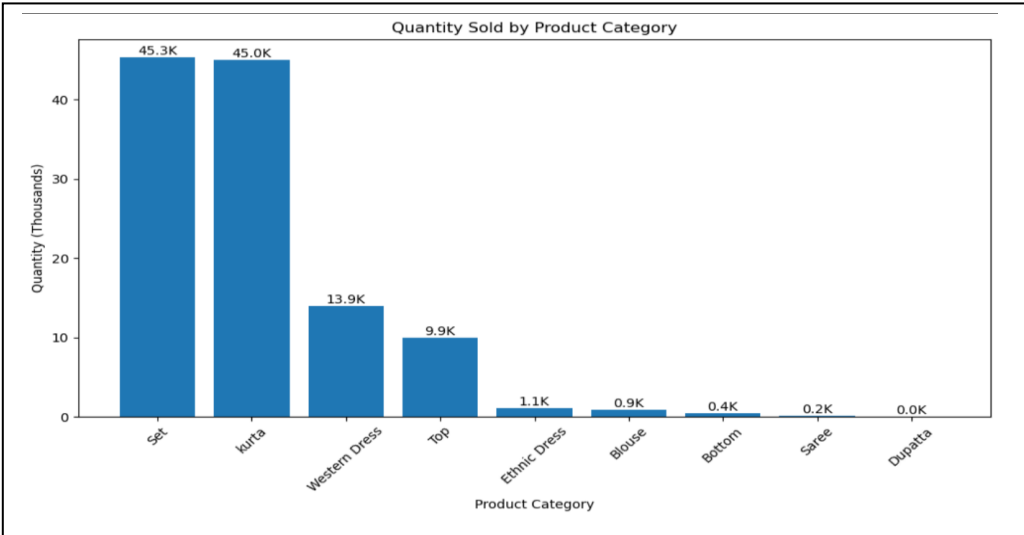
## Visualization 1: Revenue by Product Category



### Business Insights:

The revenue analysis shows that Set is the highest revenue-generating category at approximately ₹39.2 million, followed by Kurta and Western Dress. Revenue is concentrated among a few major categories, with Set contributing significantly due to its strong sales volume and higher Average Selling Price (ASP).

## Visualization 2: Quantity Sold by Product Category

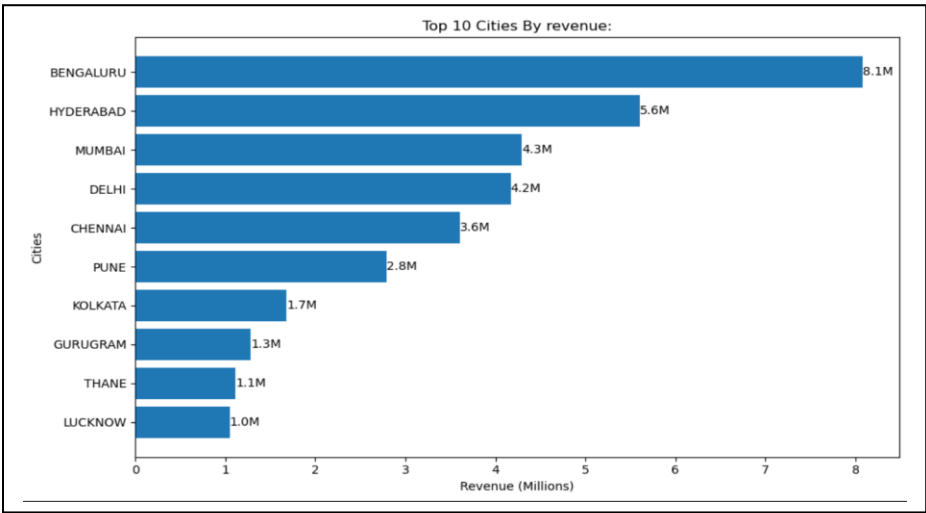


### Business Insight:

Set and Kurta recorded almost identical sales quantities, at approximately 45.3K and 45.0K units respectively. However, Set generated substantially higher revenue than Kurta. This difference is explained by Set's considerably higher average selling price of ₹865.64 compared with ₹472.85

for Kurta. Therefore, Set's revenue leadership is influenced by both high sales volume and higher selling price.

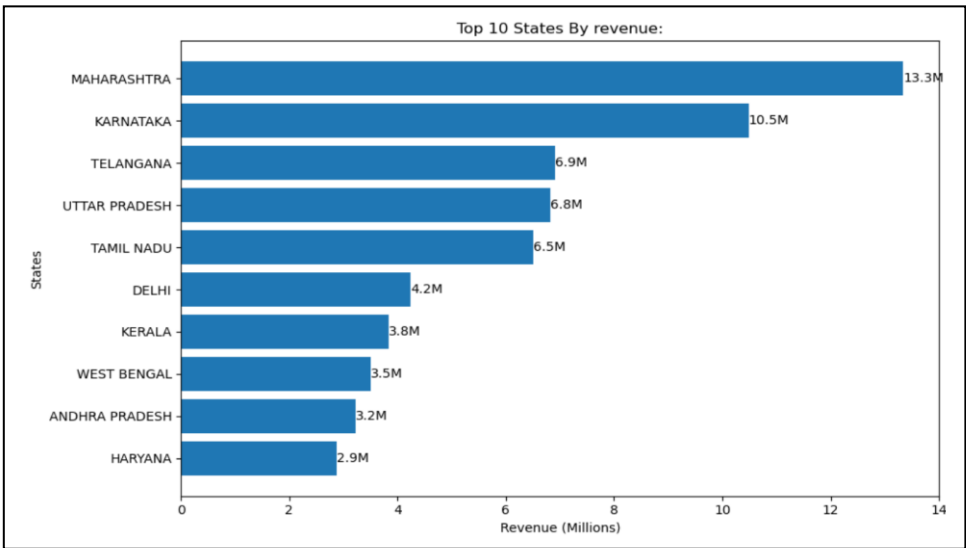
### Visualization 3: Top 10 Cities by Revenue



#### Business Insight:

The top 10 cities contribute a significant share of the recorded revenue, with Bengaluru leading at approximately ₹8.08 million, followed by Hyderabad and Mumbai. This indicates that major metropolitan cities are important markets for the business and could be prioritized for targeted marketing, customer acquisition, and expansion strategies.

### Visualization 4: Top 10 States by Revenue



### Business Insights:

The top 10 states contribute a major share of the recorded revenue, with Maharashtra leading at approximately ₹13.34 million, followed by Karnataka and Telangana. This indicates that revenue is concentrated in a few key states, highlighting these regions as important markets for targeted marketing, customer acquisition, and future business expansion.

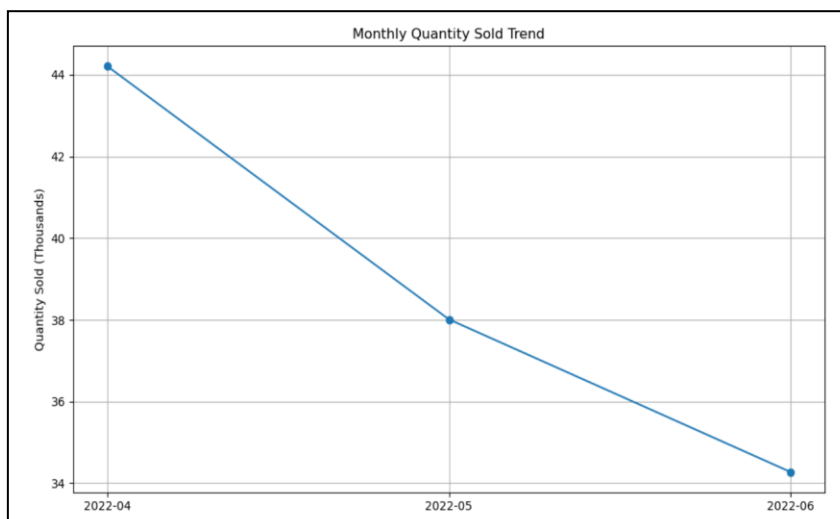
### Visualization 5 — Monthly Revenue Trend



### Business Insights:

Revenue peaked in April 2022 at approximately ₹28.84 million and declined consistently in May and June, indicating a downward revenue trend during the observed full-month period.

### Visualization 6 — Monthly Quantity Trend.



## Business Insights:

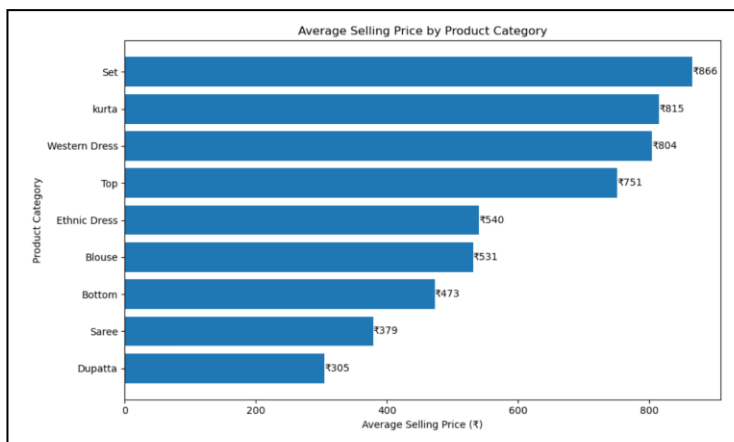
The quantity sold shows a consistent decline from April to June, falling from approximately 44.2K units in April to 34.3K units in June. This indicates a reduction in sales volume during the observed period and supports the downward revenue trend we identified earlier.

## Visualization 7- Monthly Average selling price



The **Average Selling Price (ASP)** increased from **₹652 in April to ₹690 in May** before slightly declining to **₹683 in June**. Despite this fluctuation, ASP remained relatively stable, suggesting that the decline in overall revenue was driven mainly by **lower sales quantity rather than a significant decrease in selling prices**

## Visualization 8- Category wise Average Selling Price



## Business Insights:

The ASP analysis shows that Set has the highest average selling price (ASP), followed by Saree and Western Dress, while Dupatta, Bottom, and Kurta have comparatively lower ASPs. This indicates that higher-priced categories such as Set contribute more revenue per unit, while lower-ASP categories rely more heavily on sales volume to generate revenue.

## ❖ Final Findings Review

### ➤ Finding 1 — Product Category Performance

Set is the strongest category:

- Highest revenue: ~**₹39.2M**
- Highest quantity: ~**45.3K units**
- Highest ASP among categories

**Business meaning:** Set is the most important product category and should receive attention in inventory planning, promotions, and marketing.

### ➤ Finding 2 — Revenue Concentration by Geography

- Maharashtra is the highest-revenue state at approximately ₹13.34M.
- At city level, **Bengaluru** leads with approximately **₹8.08M**, followed by Hyderabad and Mumbai.
- **Business meaning:** A significant portion of revenue comes from major markets, so these regions are important for customer acquisition and expansion strategies.

### ➤ Finding 3 — Revenue Trend

- For the comparable full-month period:  
April → May → June  
**Revenue declined:**  
April: ₹28.84M  
May: ₹26.23M  
June: ₹23.43M

- **Business meaning:** The business experienced a declining revenue trend after April and should investigate the factors behind the reduction.

➤ **Finding 4 — Quantity Trend**

- Quantity also declined:

April: **44.21K**

May: **38.01K**

June: **34.28K**

**Business meaning:** The revenue decline is strongly associated with **lower sales volume**.

➤ **Finding 5 — ASP Trend**

- ASP remained relatively stable:

April: ~₹652

May: ~₹690

June: ~₹683

➤ **The overall business story:**

The Set category is the strongest product category, while revenue is concentrated in major states and cities. However, after April 2022, both revenue and sales quantity declined, while ASP remained relatively stable. This suggests that the primary issue during the observed period was declining sales volume rather than significant pricing pressure.

❖ **Final Business Insights:**

1. **Product Category Performance:**

Set is the strongest product category, generating approximately ₹39.2M in revenue and selling around 45.3K units. Its high revenue is supported by both strong sales volume and a relatively high ASP.

2. **Geographic Concentration:**

Maharashtra is the top revenue-generating state at approximately ₹13.34M, while Bengaluru leads at the city level with approximately ₹8.08M. This highlights major urban markets as important revenue centers.

3. **Revenue Decline:**

Revenue decreased from approximately ₹28.84M in April to ₹23.43M in June, indicating a clear decline after April during the comparable period.

**4. Declining Sales Volume:**

Quantity sold decreased from approximately 44.2K units in April to 34.3K units in June. The simultaneous decline in revenue and quantity suggests that lower sales volume is a major contributor to the revenue decline.

**5. Relatively Stable Pricing:**

ASP remained comparatively stable, moving from approximately ₹652 in April to ₹690 in May and ₹683 in June. Therefore, the revenue decline does not appear to be primarily caused by falling average selling prices.

**6. Data Quality & Standardization:** Inconsistent city names such as Bangalore/Bengaluru and New Delhi/Delhi were standardized before geographic analysis. This is important because inconsistent naming can split the same market into multiple categories and distort business insights.

▪ **Summary:**

The business is strongly dependent on a few high-performing product categories and major geographic markets. While pricing remained relatively stable, both revenue and sales volume declined after April, suggesting that maintaining sales volume is a key area for business attention.

❖ **Final Project Conclusion:**

The Amazon Sales Analysis highlights strong performance from a few key product categories and geographic markets, with the Set category emerging as the leading revenue contributor. Maharashtra was the highest-revenue state, while Bengaluru was the leading city. However, revenue and sales quantity declined consistently from April to June 2022, while the Average Selling Price remained relatively stable. This suggests that the decline in revenue was primarily associated with reduced sales volume rather than significant pricing changes. The analysis also demonstrates the importance of data cleaning and standardization, particularly for geographic data, to ensure reliable business insights. Overall, the findings can help businesses identify high-performing products and markets while focusing on strategies to improve sales volume and sustain revenue growth.